

The meeting pack — what to test, what to bring, what to say, what to ask for

Two days. One application that has been walked end to end once. Below: the three questions you asked, answered; a test plan that fits in the time; the equipment; the twenty-minute walk; the questions you will get; and the one thing to ask for at the end.

01 Your three questions

Do I need a PPT, or just demonstrate the application?

Demonstrate. No slide show during the demo. Carry one printed page.

A DRM sees slides every day; a live shop-floor screen that answers a question in front of them is what they remember. Slides also invite the meeting to become a slide review. What works in that room:

- **Two minutes of framing, spoken, before you touch the laptop** — who you are, what you built, in one breath: *“A QC record for the spring section and the wagon pipeline: the inspector taps the band on the bench, the wagon carries its own checklist, wheels, photographs and parts, and the DRM sees the shop as it is — every figure counted from the record, nothing estimated. It runs on one PC in the shop with no internet.”*
- **Then the application, twenty minutes** (section 04 below), on the projector from the laptop, with the tablet for the bench and the camera.
- **A one-page leave-behind, printed** (section 09) — what it is, what it needs, what we ask. That is the "PPT" in their file. If their office insists on a deck for the record, say so and I will make a six-slide one in an hour; it is not needed for Wednesday.

Do I need an external camera, or is the laptop / tablet camera fine?

The tablet's (or phone's) rear camera is the one that matters. Bring a cheap USB webcam only as a fallback for the laptop.

- Everything the camera does on the floor — photograph a spring while sorting, read a painted wagon number, the bogie assembly frames — is done on the **tablet's rear camera**. That is what the shop will use, so that is what you show.
- The laptop's built-in camera faces *you*, not the spring on the table. If you want to show the camera on the projector, a ₹1,500 USB webcam pointed down at the table does it. Optional — the record does not depend on the camera, and the walk is written so the camera steps are on the tablet.
- What the camera needs is not a better lens; it is **https and the certificate installed on the tablet** (TABLET_TRUST.md). Without that the browser keeps the camera shut. Do that tonight, not in the room.
- Bring something to photograph: **one real coil spring** (borrow from the shop, or any outer spring) and **a sheet with the eleven-digit wagon number 22057123457 printed large** (100-pt bold — it is a valid BOXNHL number; the camera reads the stencilled digits, never the SER/BOXNHL/... form). Reading it off paper is a fair demonstration of reading it off steel.

What other information do I need?

Six things, all below: the two-day test plan, the kit list, the timed walk, the answers to the questions they will ask, the "more than you asked for" list, and the ask.

One more that is not a document: **the network in the room**. An office Wi-Fi usually blocks one device from talking to another, so the tablet may not reach the laptop on it. Plan on the laptop's own hotspot (Windows: *Settings* → *Network* → *Mobile hotspot*) or a travel router you bring, and test that exact arrangement on Tuesday. Nothing needs the internet.

Zapheit on the day: it is your product, so the rehearsal now carries the key by default (`WITHOUT_ZAPHEIT=1 bash scripts/rehearsal.sh start` leaves it out). The readiness panel shows it as configured. When the question comes, the honest line is above.

Plan C, tested: if the laptop has internet, `bash scripts/rehearsal.sh tunnel` (Mac) prints a public `https://...trycloudflare.com` address that any phone on any network can open — no certificate needed, the DRM's own phone included. It takes up to a minute to appear, it changes every start, and the data passes through Cloudflare encrypted: a demonstration tool, never for the live record. Turning the phone's mobile data *off* then stands in for "Wi-Fi off" in the offline step.

01b Your kit, exactly — MacBook Air, iPad, Pixel hotspot

There is no Windows laptop and therefore **no START.cmd** on the day. START.cmd is the shop PC's starter; on the Mac the same thing is one command. Here is the whole setup, in order:

1. **Pixel:** turn on *Mobile hotspot* (Settings → Network & internet → Hotspot & tethering). Leave mobile data on — the hotspot needs it.
2. **Mac:** join the Pixel's hotspot Wi-Fi. **iPad:** join it too.
3. **Mac, Terminal:** `cd ~/Desktop/WRS_Raipur` then `bash scripts/rehearsal.sh start`. Two to three minutes the first time. It reissues the server certificate for the address the hotspot gave the Mac — that is why the hotspot goes on *first*.
4. **Mac:** `bash scripts/rehearsal.sh status` prints three addresses: `https://localhost:3200` for the projector; `https://10.x.x.x:3200` (the number changes with every network); and **`https://pratiks-macbook-air-3.local:3200` — use this one on the iPad**, it is the same in Kolkata, Raipur, on the hotspot and on the shop Wi-Fi, so the home-screen icon keeps working.
5. **iPad and Pixel, once ever:** install the workshop CA — the file `.rehearsal-certs/lan-cert.crt` in the project folder (AirDrop it to the iPad; email or cable to the Pixel). iPad: Settings → Profile Downloaded → Install, **then** Settings → General → About → Certificate Trust Settings → switch on *WRS Raipur workshop CA*. Pixel: Settings → Security → Encryption & credentials → Install a certificate → CA certificate. After this the address opens with no warning on any hotspot or Wi-Fi, forever — the certificate no longer depends on the address.
6. **iPad:** open `https://pratiks-macbook-air-3.local:3200` in Safari → Share → Add to Home Screen. Sign in as inspector1. *Sorting* → *Photograph springs while sorting*: the camera opens. If it does not, the CA trust switch (step 5) is off.
7. **The Pixel cannot open the address itself** — a phone does not route to the devices on its own hotspot (tested 21 Sep). It is the hotspot and nothing else. **The iPad scans the certificate QR** (3.8 / 4.9): the iPad's Camera app → the yellow link → Safari → VERIFIED. If a second phone is wanted for scanning, it joins the hotspot like the iPad.
8. **Projector:** HDMI to the Mac, mirror the display, Chrome at `https://localhost:3200`, zoom 110–125 %.

Test this exact arrangement on Tuesday, hotspot and all. If the Mac's number changes on Wednesday — a different city, a different network — nothing breaks: the server reissues its own certificate within thirty seconds of any change, the iPad trusts the CA once, and the `.local` address never changes.

The internet in the room: not needed by anything. The Pixel's data is only there so the hotspot exists. **Plan C** if the iPad refuses the hotspot: `bash scripts/rehearsal.sh tunnel` gives a public address any device can open (needs the Mac to have internet, i.e. the hotspot).

02 Two days — the test plan

You have tested everything once. "Multiple times" is right, but the runs should differ from each other, not repeat: one run by hand for correctness, one on the real hardware for the setup, and the demo walk itself rehearsed until it is boring. Tick as you go.

Today, Monday 21

correctness — on the Mac and your phone

- ☐ **Run 1 — the 70-step walkthrough by hand**, all four roles, from the artifact (`bash scripts/rehearsal.sh start` ; phone on Wi-Fi, mobile data off, certificate installed). Use the *Inspector walk* page for section 2 — it has the screenshot for every step.
- ☐ Press **Show summary**, send me the "did not match" lines. I fix, the gate runs, I push. Same evening.
- ☐ Section 6 (stop / log / backup / restore) once, exactly as written — it is five commands.
- ☐ Get the shop's spring and the printed wagon number ready for Tuesday. Charge the tablet.

Tuesday 22

the real arrangement — MacBook Air on the Pixel hotspot, iPad, Pixel

- ☐ Section 01b, steps 1–8, exactly as they will be on Wednesday: Pixel hotspot first, Mac and iPad on it, `start`, CA installed on the iPad and the Pixel, camera opens on the iPad.
- ☐ `bash scripts/preflight.sh` green on the Mac (sixteen steps).
- ☐ Copy the stick anyway — `dist-shop/wrs-raipur` + the Node 22 installer — as the second copy of the demo and the thing the shop will install (READ-ME-FIRST.txt).
- ☐ **Run 2 — the walkthrough again, on this arrangement**, at least sections 1, 2, 3 and 6 with the iPad in hand. This is the run that finds hotspot- and iPad-only faults.
- ☐ **Run 3, 4, 5 — the twenty-minute walk (section 04) three times, timed**, out loud, with the tablet in hand. By the third run you should not be looking for buttons.
- ☐ Take a backup (section 6.4). Put the stick, the printed page, the spring and the number sheet in the bag. Charge everything.

Wednesday 23

the room

- ☐ Arrive early. Pixel hotspot on → Mac and iPad on it → `bash scripts/rehearsal.sh start` → `status`. Sign in as drm1: the tiles under *Shop Floor — Right Now* show figures, not "Not yet known".
- ☐ Projector: `https://localhost:3200`, no warning. iPad: the hotspot address, camera opens.
- ☐ Do not re-seed on the day. The record you rehearsed on is the record you show.
- ☐ Framing (2 min) → walk (20 min) → questions → the ask (section 07). Leave the page.

Not before Wednesday: no new features, no wording changes you have not rehearsed with. From Tuesday evening the only change allowed is a fix for something you saw fail on the laptop.

03 What to carry

Must

- The MacBook Air, rehearsed, charger, **USB-C → HDMI adapter** (and VGA if the office is older).
- The iPad with the CA installed and the app on its home screen; the Pixel with the CA installed; chargers for both.
- The USB stick: bundle + Node installer (the second copy of the demo, and what the shop installs).
- The Pixel is the hotspot. A power bank for it — a hotspot drains a phone in two hours.
- One real coil spring; a sheet with **22057123457** printed large (the eleven-digit form the camera reads).
- **Print, from docs/handouts/:** two copies of *drm-dossier.pdf* (16 pages — the full document: roles, security, logs, photographs and storage, learning, the shadow run; one for the DRM's file, one for the CWM); six copies of *drm-handout.pdf* (3 pages — the short version, for everyone else in the room); one copy each of *inspector-walk.pdf* (30 pp), *supervisor-walk.pdf* (22), *drm-walk.pdf* (14), *admin-walk.pdf* (15) — the step-by-step with pictures, to leave with the shop. Nothing in them says how they were made.
- The shop's G-95 Rev-II pages (Tables 28–33) — the source of the six bands, in case a hand goes up.

Useful

- A USB webcam for the laptop, if you want the camera on the projector.
- The printed, ticked walkthrough — proof it was tested by hand, dated.
- A printed certificate with its QR (from a released wagon's certificate page) — let someone scan it with their own phone: it verifies with no server.
- Backup key file, on a separate stick, kept in your pocket, not the laptop bag.
- A power strip.

04 The walk — twenty minutes, in this order

Shop floor first, dashboards second, proofs last. Say the sentence in italics before each screen; it tells the room what to look for.

MIN	WHERE	DO	SAY
0–2	Standing	Nothing yet.	<i>"A QC record for springs and wagons, on one PC in the shop, no internet. Every figure you will see is counted from the record — nothing is estimated, and where the evidence is thin the screen says so."</i>
2–6	Tablet · inspector1 · Sorting	Tap three bands. Undo one. Tick photographing, hold up the spring, tap its band; hand in front → <i>person excluded</i> . Show <i>Your record today</i> . Condemn one: Crack .	<i>"One tap per spring — the six bands of your own G-95, Tables 28 to 33. The gauge is on every record. The camera keeps the photograph; it does not decide."</i>
6–9	Tablet · A wagon · WR/BCNHL/40112 · Photos	Four frames. Bogie 1·A agree. Bogie 2·B: "a blind recount is needed" — no figures. Open the counter: no expected number.	<i>"The second person counts blind. The inspector is never told what the count should be."</i>
9–12	Laptop · supervisor1 · same wagon · Photos → Release checks	Supervisor sees "6 of 7 outer". Release checks: the short count waits as an advisory, by name. Then SER/BOXNHL/30914: blocked by a condemned spring and the axle-3 pair at $917.5 < 919$ mm. Checklist → wheels: the chalk figure, the limit and the drawing beside it.	<i>"Nothing leaves the shop with a known short pocket or a wheel under the last-shop-issue diameter — and the drawing number is on the screen, not in someone's memory."</i>
12–14	Laptop · supervisor1 · Ask the Records	<i>which wagon type condemns the most snubbers this quarter · is any gauge reading high</i> . Open <i>How this was computed</i> .	<i>"The two outer gauges disagree by 1.5 mm. With two gauges it cannot say which is off, so it says both go against the master. It refuses to guess — that is the point."</i>
14–17	Laptop · drm1 · Dashboard → Spring analytics	When does today's pile finish · how many bogies can we build · where wagons wait · what keeps coming back. Then the standard report.	<i>"Every figure carries its n. This distribution of your springs against the standard is something RDSO has never had from a shop."</i>

MIN	WHERE	DO	SAY
17–19	Laptop · drm1 · Audit → certificate	Verify chain. Open a released wagon's certificate; scan its QR with the iPad's Camera app (or the DRM's own phone once it is on the hotspot): <i>verify.html</i> says VERIFIED with no server.	<i>"The record proves it was not altered after it was written. The certificate proves itself anywhere, with no login."</i>
19–20	Laptop · supervisor1 · Shadow run	The week's log; the one case the register passed and the app condemned.	<i>"This is how we propose to start: two weeks beside the register, both recording, and this screen decides go-live."</i>

Skip 12–14 if time is short; never skip 2–6 or 17–19.

05 Questions you will get – and the honest answers

- **"Does the camera decide?"** No. The camera photographs and counts; a person decides. It may only auto-sort once blind reads on this shop's own springs have earned it — and the server refuses it until then. Today the count of labelled photographs is zero, and the screen says so.
- **"Where do these percentages come from?"** Every rate has a count under it. There is no confidence score anywhere that a person did not enter or the record did not count.
- **"Does anything leave the shop?"** The record never does: one PC, one database file, the photographs, tablets on the shop Wi-Fi — no cloud, no internet needed. One thing is theirs to switch on: **Zapheit**, your own service, which when a key is set receives manual questions, spoken sentences and the shift's facts so a model can *find* the right clause or field — it never produces a number, and with the key removed the app does everything it did before, on its own. Say it as a choice they make, not a dependency.
- **"What if the PC dies / the Wi-Fi drops?"** `START.cmd` restarts the server and keeps a dated log; the tablet keeps working offline and sends later without duplicating; the backup is encrypted, on another disk, and a restore onto a second machine has been drilled (`RESTORE_DRILL.md`).
- **"What hardware?"** One ordinary Windows PC in the spring section (any PC from the last five years), two Android tablets (₹15–20k each, with a rear camera), the shop's existing Wi-Fi or one access point in the bay. No server room.
- **"Who maintains it?"** Two `.cmd` files for the shop: `START` and `backup`. The admin screen manages accounts, checklist rules and gauges without a programmer.
- **"Is this the model's number?"** No model produces a number. Ask the Records lets a model pick *which* question; the arithmetic is fixed SQL, and the rows are shown.
- **"What is not proven yet?"** Say it before they ask: the camera on real springs, the pocket count on real bogie photographs, the bearing sound on a real bad CTRB, and the shadow-run verdict. All four wait for the shop floor — which is what the pilot is for.
- **"What does it cost?"** Give the number you have decided. Do not improvise one in the room.
- **"Can it work with CRIS / the existing MIS?"** The wagon passport is a signed, portable file any system can import; the certificate QR verifies anywhere. Nothing here needs CRIS to change; an export to their format is a report, not a rebuild.

06 Beyond what was asked — say these only if the walk has gone well

- **A certificate that proves itself anywhere** — the QR, no server, no login. And a signed wagon passport the next POH shop can import.
- **The shop measures the standard** — the free-height distribution of this shop's springs against the G-95 bands, with n on every figure: evidence RDSO has never had from a shop.
- **Ask the Records with the query shown** — questions in plain words, arithmetic in fixed SQL, every number linked to its rows.
- **Gauge drift from the record itself** — a gauge reading high shows up in the distribution without a calibration visit.
- **The wagon that will miss its date** — computed from stage dwell, not predicted. (Ready to show on the dashboard after the meeting.)
- **Hindi on every screen** the inspector holds.

07 The ask — one sentence, at the end

"We ask for a two-week shadow run in the spring section: one PC, two tablets, two inspectors and one supervisor, recording beside the register as they do today. At the end the shadow-run screen shows where the two disagreed and who was right, and that decides go-live."

Then the three things you need from them to start: a PC in the bay with power and Wi-Fi; the names of the two inspectors and the supervisor (real accounts are created on day one, demo accounts deleted); the shop's G-95 Rev-II and gauge register to check against.

Do not ask for a decision on production in the room. Ask for the shadow run; production follows the evidence.

08 If something breaks in the room

- **Screen blank / "cannot reach":** is the START.cmd window still open on the laptop? Double-click it; ten seconds.
- **Tablet says not secure or camera will not open:** the certificate is not trusted on that tablet. Finish the walk on the laptop; the record does not depend on the camera.
- **Tablet cannot reach the laptop:** office Wi-Fi is isolating devices — switch to the laptop hotspot you tested Tuesday.
- **A tile says "Not yet known":** that is the system refusing to guess. Say so; it is the point.
- **Anything else:** `logs\wrs-<today>.log` has the reason; carry on with the next screen.

09 The one-page leave-behind (print this section)

WRS Raipur — Spring & Wagon QC

A quality record for the spring section and the wagon pipeline · runs on one PC in the shop · no internet

What it does	Sorts springs by the six G-95 bands with one tap per spring, with the gauge on every record. Carries each wagon's checklist, wheel readings, photographs, parts ledger, air-brake proforma and stage history from entry to release. Blocks release on a condemned spring, a short pocket count or a wheel below the last-shop-issue diameter, and says why.
What the DRM sees	When today's pile finishes; how many bogies can be built from stock; where wagons wait; what keeps coming back — every figure counted from the record with its n. The shop's springs against the standard.
What it proves	A hash-chained audit record; a release certificate whose QR verifies on any phone with no server; a signed wagon passport for the next shop.
What decides	A person, always. The camera photographs and counts; it may only sort on its own after blind reads on this shop's springs have earned it, and the server refuses it until then.
What it needs	One Windows PC in the bay; two Android tablets with rear cameras; the shop Wi-Fi or one access point. Two .cmd files to run; backups encrypted to a second disk.
Not yet proven	The camera on real springs, the pocket count on real photographs, the bearing sound, the shadow-run verdict — all four wait for the shop floor.
Proposed start	A two-week shadow run in the spring section beside the register: one PC, two tablets, two inspectors, one supervisor. The disagreement log decides go-live.
Contact	Pratik Agarwal · pratik96agarwal@yahoo.com

Companion pages (yours, not for the room): the 70-step [Walkthrough test script](#) (tick page); the walks with the machine's screenshots — [Inspector](#), [Supervisor](#), [DRM](#), [Administrator](#). For the room: [docs/handouts/](#) (PDF and HTML, nothing about the tooling). In the repo: docs/DEMO_DAY.md, docs/TABLET_TRUST.md, docs/RESTORE_DRILL.md.